Tutorial No : AMVS 2

Topic : Air-conditioning Equipment

1. State the difference between the construction of the open, hermetic and serviceable hermetic compressors. [3 points]

2. Name the types of compressors by method of compression. [2 points]

3. Describe, with the aid of diagram(s), the construction and operation of the following metering devices:

   (a) Thermostatic expansion valve [10 points]
   (b) Low side float [8 points]

4. Sketch a labeled diagram of a 2-pass shell and tube condenser with indications of the directions of flow of the different fluids. [6 points]

   Explain why it is important to keep the water side of the condenser tubes clean and free of obstructions. [5 points]

5. State the function (s) of the following accessories and indicate on a sketch of the vapour compression refrigeration circuit, the appropriate locations for each of the accessories: [12 points]

   (a) High pressure cut-out
   (b) Low pressure cut-out
   (c) Filter drier
   (d) Sight glass

6. Sketch a labelled diagram of a draw-through air handling unit showing the following components: [6 points]

   Mixing box, cooling coil, air filters, condensate drain pan and pipe, AHU fan; supply duct, return duct and outdoor air duct connections; and chilled water supply and return pipe connections

7. Differentiate between a draw-through and a blow-through air handling unit. Discuss the effects of dirty filters on the performance of an air handling unit. [5 points]

8. During an energy audit, the following data were recorded for equipment shown in Figure 8:-

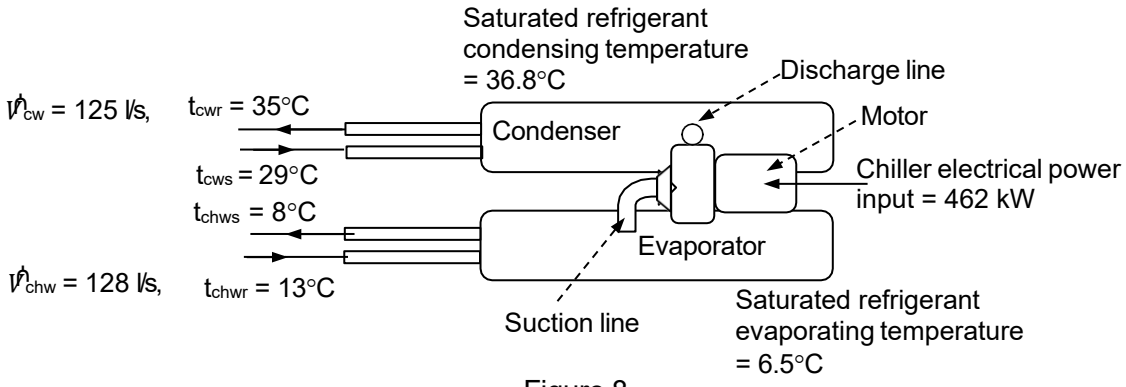


Figure 8

Name the equipment shown in Figure 8 and calculate the following: -

(a) cooling capacity of the chiller in kW. [2 points]
(b) chiller Electrical Power Input/RT. [2 points]
(c) COP. [1 point]
(d) chiller heat rejection rate. [2 points]
(e) evaporator approach. [1 point]
(f) condenser approach. [1 point]

9. Automatic condenser tube cleaning systems are sometimes installed in water-cooled chillers. Name the two (2) types of automatic cleaning system are that commonly used and describe the operation of any one type. [9 points]

10. Arrange the starting sequence for the following equipment of a water-cooled chilled water system in chronological order. [2 points]

Water-cooled chillers
Chilled water pumps
Condenser water pumps
Cooling Towers

11. Explain the reason(s) for installing chilled water flow switches and condenser water flow switches in the pipework of the chilled water circuit and the condenser water circuit respectively. [12 points]

12. Define the terms "range" and "approach" as applied in cooling towers.

Data:

Required cooling capacity for a chiller	: 600 kW
Design chilled water supply temperature	: 8°C
Design chilled water return temperature	: 15°C
Outside design condition	: 32°C dbt, 26°C wbt
Condenser heat rejection factor	: 1.25
Design cooling tower approach and range	: 3°C and 6°C respectively

Determine the following using the data given above:

(a) Required chilled water flow rate for the chiller. [2 points]
(b) Required cooling tower heat rejection capacity. [1 point]
(c) Temperatures of the water entering and leaving the cooling tower. [2 points]

13. Sketch a labeled diagram of a cooling tower and describe the function(s) of the following components of a cooling tower:

(a) Drift eliminators [5 points]
(b) Fill or packing [4 points]

14. Explain why it is necessary to supply make-up water to cooling towers. [5 points]

15. Briefly explain the term “bleed-off” in the context of cooling tower operation.

16. List six (6) maintenance tasks which are required to be carried out for the following equipment:

(a) air-cooled split unit [6 points]
(b) water-cooled packaged unit [6 points]
(c) air handling unit [6 points]
(d) cooling tower [6 points]